

NOAH ZHANG

(+1) 404-703-7455 | zhangnuoya969@gmail.com | Atlanta, GA
[LinkedIn](#)

EDUCATION

Georgia Institute of Technology

Ph.D. in Physics (Advisor: Surabhi Sachdev)

Atlanta, GA

Aug 2023 – Present (Exp. 2028)

- Specialization: Gravitational Wave Detection, GstLAL Pipeline, Low-Latency Signal Processing.

Georgia Institute of Technology

M.S. in Computational Science and Engineering (CSE)

Atlanta, GA

Aug 2023 – Exp. 2025

- **Relevant Coursework:** High Performance Computing, Computational Data Analysis, Modeling & Simulation, Algorithms.

Nanyang Technological University (NTU)

B.Sc. in Physics (First Class Honors)

Singapore

Aug 2019 – May 2023

- **GPA: 4.62/5.00** (Major GPA: 5.00/5.00). Awarded Full Ministry of Education Scholarship.

TECHNICAL SKILLS

Languages/Libraries: C++ (STL, Boost), Python (NumPy, SciPy, Pandas, **ccxt**, **pykalman**), SQL, $\LaTeX$

Computation: High Performance Computing (HPC), MPI, HTCondor, Git, Linux/Unix

Quantitative: Monte Carlo Simulation (MCMC), Bayesian Inference, Time-Series Analysis, MLE, Signal Processing

RESEARCH & QUANTITATIVE EXPERIENCE

LIGO Scientific Collaboration (Georgia Tech)

Aug 2023 – Present

Graduate Research Assistant – Low-Latency Signal Processing

Atlanta, GA

- **Search Algorithm Optimization:** Co-architected the “Targeted Search” method for GstLAL by integrating Electromagnetic (EM) counterpart data as Bayesian priors. Improved signal detection sensitivity by **3 orders of magnitude (1000x)** while maintaining the same latency profile.
- **Production Analysis Lead:** Led the Intermediate Mass Black Hole (IMBH) offline search for the GWTC-5 production run. Refined background noise estimation models, achieving a **13% improvement** in noise rejection accuracy over the previous baseline.
- **Statistical Validation:** Conducted parameter estimation (PE) review for the SGNL pipeline. Optimized statistical estimators, improving P-P plot (probability-probability) consistency and result reliability by **33%**.
- **Pipeline Engineering:** Maintain the GstLAL hybrid **C++/Python** pipeline, processing terabytes of streaming data with sub-second latency for real-time trigger generation.

Nanyang Technological University

May 2020 – Aug 2022

Undergraduate Researcher – Stochastic Modeling & Simulation

Singapore

- **Algorithmic Speedup:** First author of *Dynamics, Statistics, and Task Allocation of Foraging Ants*. Optimized the agent-based simulation by combining MCMC with a customized Random Walk algorithm.
- **Performance Gain:** Reduced simulation runtime from **days to hours (~50x speedup)** through code optimization and efficient sampling strategies.

SELECTED QUANTITATIVE PROJECTS

High-Frequency Market Microstructure Signal Extraction | Python, Kalman Filter, MLE

Aug 2025 – Present

- **State-Space Modeling:** Developed a **Kalman Filter** to separate efficient price signals from microstructure noise (bid-ask bounce) in high-frequency Bitcoin tick data, modeling price evolution as a random walk with mean-reverting error.
- **Constrained Optimization:** Implemented **Maximum Likelihood Estimation (MLE)** to calibrate process/measurement noise matrices (Q and R). Diagnosed and resolved regime-switching overfitting by anchoring measurement noise to empirical tick-size volatility.
- **Benchmarking:** Validated custom manual implementation against the **EM algorithm** (via *pykalman*), demonstrating superior signal retention in high-volatility regimes.

LEADERSHIP & AWARDS

PLANCKS Physics Competition (Singapore)

Mar 2022 – May 2022

Silver Medalist (2nd Place)

Singapore

- Collaborated in a team of four to solve complex theoretical physics problems under strict time constraints.

Odyssey Research Symposium | Best Presentation Award

Dec 2020

- Awarded for excellence in technical communication and data presentation.